

Santiago De Jesus Villalobos Gonzalez

Quantitative Researcher & AI/ML Data Ops Specialist

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LinkedIn: linkedin.com/in/prefrontalcorporate • **GitHub:** github.com/SVG-campus • **Kaggle:** kaggle.com/seasthaalores/code

PROFILE SUMMARY

Quantitative Researcher, Apple alumnus, and Solo Founder specializing in Topological Data Analysis (TDA), Multiscale Entanglement Renormalization (MERA-KMPA), and persistent homology ($\beta_0, \beta_1, \beta_2$). Ex-Apple AI/ML Data Ops Associate with proven expertise in high-volume training data pipelines, PyTorch, C++, and systematic trading engines. Collaborating with Stanford Emeritus Prof. Gunnar Carlsson on non-stationary financial data manifolds and verifier-backed causal AI reasoning.

QUANTITATIVE RESEARCH & AI BREAKTHROUGHS

AetherQuant Quantitative Trading Engine & MERA-KMPA Framework | Founder & Lead Researcher

- **Persistent Homology Trading Signals:** Developed a proprietary trading overlay mapping financial point clouds onto complex projective manifolds ($\mathbb{CP}^n$), tracking Betti loop persistence (β_1) to detect market regime shifts prior to price volatility spikes.
- **Proven Performance:** Demonstrated +12.0% after-market gains and +8.0% live execution bursts in paper/prop testing; presented algorithms to former CEOs of AMEX and Weight Watchers.

Upgraded-7B-Full-Universe-Swarm & Causal Topology Benchmark | Creator

- **100% Verification Accuracy:** Engineered a local 7B quantized model integrated with a FastMCP MERA-KMPA symbolic solver, achieving 100.0% pass rate (50/50 tasks) across formal topological and d-separation suites.
- **Locked Gate Remediation:** Built and locked frozen_v2 (SHA-256 locked); demonstrated targeted remediation outside the suite, raising unaided model-only reasoning from 41.2% → 47.1% → 58.8% with zero benchmark drift.

WorldQuant International Quant Championship (IQC 2026) | Quant Researcher

- **Top 20% Global Ranking:** Ranked in the Top 20% globally in Stage 1 of the 2026 WorldQuant IQC, constructing market-neutral, medium-frequency alpha signals.

WORK EXPERIENCE

New York Life | Registered Agent & Investment Venture Lead (Santa Rosa, CA | July 2026 – Present)

- Advise UHNW single-family trusts in Northern California on risk management and asset allocations.
- Structuring subsidiary investment and life insurance vehicles under New York Life tutelage.

Apple | AI/ML Data Ops Associate (Cupertino, CA | Oct 2024 – Dec 2025)

- **Pipeline Optimization:** Engineered patent-pending data operations architecture optimizing high-volume training data streams, eliminating pipeline bottlenecks for Siri project scaling.
- **Data Integrity at Scale:** Managed high-volume data streams for machine learning model training, ensuring data integrity across cross-functional engineering teams.

The Grind Business Pitch Competition | Finalist & Presenter (New York City, NY | Apr 2025 – Sep 2025)

- Selected as finalist in NYC to pitch high-stakes quantitative financial models before veteran business executives and venture investors.

Stanford Medicine | L.E.A.P. Research Fellow (Stanford, CA | Aug 2024 – May 2025)

- Directed research initiatives within the Applied Medical Leadership Program.

EDUCATION

Syracuse University | B.A. Computer Science & B.S. Applied Mathematics (GPA: 3.9)

Campus, Inc. | A.A. Business Administration & Minor in Applied AI (GPA: 4.0)

TECHNICAL SKILLS & CERTIFICATIONS

Languages & Frameworks: Python, C++, PyTorch, NumPy, SciPy, Ripser/GUDHI (TDA), SQL, Tableau, Docker, FastMCP, WebSockets, FastAPI, PostgreSQL.

Certifications: Google Cybersecurity (CXG1PRW5WWMH), Coursera Financial Markets (O5AONZY9D4), Google AI (S7OX1TNZY9YI), AI Infrastructure & Ops (Y0RCUEWASLP1), Applied Entrepreneurship (Campus, Inc.).

Awards & Honors: WorldQuant IQC Top 20% Team (2026), Debate Club Best Overall Debater (2025), President's List, Dean's List.

Languages: English & Spanish (Native Fluency).